

# Forensic Detection Report

Report: SYNTHETIC LAYOUT SAMPLE - NOT EVIDENCE  
Generated (UTC): 2026-09-10T13:34:53+00:00

This is a system-generated report. All detections and validation results are subject to human analyst review and should be treated as reviewable observations, not conclusive findings.

## Video Metadata

|                     |                             |
|---------------------|-----------------------------|
| Video filename      | SYNTHETIC_LAYOUT_SAMPLE.mp4 |
| Resolution          | 240 x 144 pixels            |
| Frame rate          | 5 FPS                       |
| Duration            | 6.000 seconds               |
| Source frame count  | 30                          |
| Camera identifier   | TEST-CAM01                  |
| Recording date/time | Not recorded                |

## Video and Detection Information

|                          |                                               |
|--------------------------|-----------------------------------------------|
| Model used               | SYNTHETIC_MODEL_REFERENCE.pth                 |
| Run status               | Completed saved run                           |
| Confidence threshold     | 50%                                           |
| Frames analyzed          | 30                                            |
| Processing device / time | CPU / 1.00 seconds                            |
| Analysis completed (UTC) | Not recorded                                  |
| Detection summary        | 4 observations; 4 positive frames; Handgun: 4 |
| Analyst review coverage  | 0 of 4 observations reviewed                  |

## Interpretation

The model generated the listed handgun and knife observations at the configured confidence threshold. A confidence score describes model certainty and is not an analyst decision. Analyst decisions document a later human assessment and do not replace or delete the original model observation.

Frame numbers are zero-based. Times are video-relative offsets and are not recording dates. Bounding boxes use [x1, y1, x2, y2] source-frame pixel coordinates. Counts represent frame observations, so the same physical object may appear more than once.

## Object Detection Observations and Reviews

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00001

|                           |                                  |
|---------------------------|----------------------------------|
| Detection                 | Handgun   Frame 0   0.00 seconds |
| Confidence / bounding box | 80.00% / [20, 30, 80, 100]       |
| Analyst Review Decision   | Not reviewed                     |
| Review timestamp (UTC)    | Not recorded                     |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00002

|                           |                                  |
|---------------------------|----------------------------------|
| Detection                 | Handgun   Frame 1   0.20 seconds |
| Confidence / bounding box | 80.00% / [20, 30, 80, 100]       |
| Analyst Review Decision   | Not reviewed                     |
| Review timestamp (UTC)    | Not recorded                     |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00003

|                           |                                   |
|---------------------------|-----------------------------------|
| Detection                 | Handgun   Frame 20   4.00 seconds |
| Confidence / bounding box | 80.00% / [20, 30, 80, 100]        |
| Analyst Review Decision   | Not reviewed                      |
| Review timestamp (UTC)    | Not recorded                      |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00004

|                           |                                   |
|---------------------------|-----------------------------------|
| Detection                 | Handgun   Frame 21   4.20 seconds |
| Confidence / bounding box | 80.00% / [20, 30, 80, 100]        |
| Analyst Review Decision   | Not reviewed                      |
| Review timestamp (UTC)    | Not recorded                      |

## TCR Information

|                                |                                          |
|--------------------------------|------------------------------------------|
| Metric                         | Temporal Consistency Rate (TCR)          |
| Result                         | 50.00%                                   |
| Interpretation                 | Calculated from saved pipeline statuses. |
| Supported observations (N_TS)  | 2                                        |
| Isolated observations          | 1                                        |
| Interruptions                  | 1                                        |
| Boundary observations excluded | 2                                        |
| Eligible total (N_TE)          | 4                                        |

## MCCR Information

|                |                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| Metric         | Multi-Camera Corroboration Rate (MCCR)                                                                                     |
| Result         | N/A                                                                                                                        |
| Interpretation | Requires two camera feeds with camera identifiers and comparable timestamps. This run has fewer than two recorded cameras. |

## Analyst Review Information

|                 |                              |
|-----------------|------------------------------|
| Review coverage | 0 of 4 observations reviewed |
| Decision totals | Not reviewed: 4              |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00001

|                         |                    |
|-------------------------|--------------------|
| Analyst Review Decision | Not reviewed       |
| Analyst notes           | No notes provided. |
| Review timestamp (UTC)  | Not recorded       |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00002

|                         |                    |
|-------------------------|--------------------|
| Analyst Review Decision | Not reviewed       |
| Analyst notes           | No notes provided. |
| Review timestamp (UTC)  | Not recorded       |

### Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00003

|                         |                    |
|-------------------------|--------------------|
| Analyst Review Decision | Not reviewed       |
| Analyst notes           | No notes provided. |
| Review timestamp (UTC)  | Not recorded       |

## Observation 5df6ab1067f8e8336fd067555822b0b3-OBS-00004

|                         |                    |
|-------------------------|--------------------|
| Analyst Review Decision | Not reviewed       |
| Analyst notes           | No notes provided. |
| Review timestamp (UTC)  | Not recorded       |

## Source References

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Source video              | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\tmp\pdfs\metrics-layout\SYNTHETIC_LAYOUT_SAMPLE.mp4 |
| Model                     | SYNTHETIC_MODEL_REFERENCE.pth                                                                                    |
| Annotated video           | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\tmp\pdfs\metrics-layout\annotated.mp4               |
| Saved summary             | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\tmp\pdfs\metrics-layout\summary.json                |
| Pipeline detections       | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\tmp\pdfs\metrics-layout\detections.csv              |
| Metric input records      | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\tmp\pdfs\metrics-layout\metric_input.csv            |
| Metric calculation script | C:\Users\howar\Documents\GitHub\cctv-weapon-detection-system\benchmarks\calculate_tcr_and_mccr.py                |

## Traceability Report

|                                 |                                                                  |
|---------------------------------|------------------------------------------------------------------|
| Report identifier               | SYNTHETIC LAYOUT SAMPLE - NOT EVIDENCE                           |
| Generated (UTC)                 | 2026-09-10T13:34:53+00:00                                        |
| Source video fingerprint        | Unavailable                                                      |
| Source video check              | unavailable                                                      |
| Saved summary fingerprint       | 5df6ab1067f8e8336fd067555822b0b3f321ab9fcd38155994b5240c8b9ef04a |
| Saved summary check             | fingerprinted_at_report_generation                               |
| Pipeline detections fingerprint | 6e59e45d6579e692173c09ccece034a6ef29ce8b98087dd6ec5b2c78b918b635 |
| Pipeline detections check       | fingerprinted_at_report_generation                               |
| Metric input fingerprint        | 575ad70a141d4971ee46feb9778d72b99f7c69a71baf48a1f447a1e3e175814b |
| Metric input check              | fingerprinted_at_report_generation                               |
| Metric script fingerprint       | ff4ab2de2e7a1d7e06c15e1fc6dc087e442f9559d61e99767860cb2503a28c1b |

SHA-256 values identify files read when this report was generated; no ingestion-time hash or chain-of-custody history is asserted.